

ISOMORPHISM OF GRAPHS

We often need to know that whether it is possible to draw two graphs in the same way. For instance, in Chemistry graphs are used to model Compounds. Different Compounds can have the same molecular formula but can differ in structure. Such Compounds will be represented by graphs that cannot be drawn in the same way. The graphs representing previously known Compounds can be used to determine whether a supposedly new compound has been studied before.

There is a useful terminology for graphs with the same structure.

Definition:

The simple graph $G=(U, E_1)$ and $H=(V, E_2)$ are isomorphic if there exists one-to-one and on-to function f from U to V with the property that a and b are adjacent in G iff $f(a)$ and $f(b)$ are adjacent in H , for all a and b in U . Such a function f is called isomorphism. (iso means equal, morphe means form)

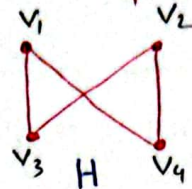
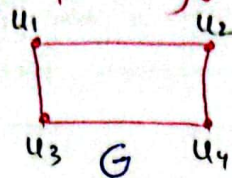
In other words

when two simple graphs are isomorphic, there is one-to-one correspondence between vertices of the two graphs that preserves the adjacency relationship.

Isomorphism of simple graphs is an equivalence relation.

Example #1

Show that the following graphs are isomorphic.



Sol:

- Both have same # of vertices and same # of edges.

$$\# \text{ of vertices : } \begin{array}{cc} G & H \\ 4 & 4 \end{array}$$

$$\# \text{ of edges : } \begin{array}{cc} 4 & 4 \end{array}$$

- Both have same # of vertices of degree d.

Degree	# of vertices	
	G	H
0	0	0
1	0	0
2	4	4
3	0	0

- one-to-one correspondence between vertices of the two graphs that preserves the adjacency relationship.

$$f(u_1) = v_1$$

$$\deg(u_1) = 2 \begin{array}{l} 2 \\ 2 \end{array}$$

$$1. \deg(v_1) = 2 \begin{array}{l} 2 \\ 2 \end{array}$$

$$2. \deg(v_2) = 2 \begin{array}{l} 2 \\ 2 \end{array}$$

$$3. \deg(v_3) = 2 \begin{array}{l} 2 \\ 2 \end{array}$$

$$4. \deg(v_4) = 2 \begin{array}{l} 2 \\ 2 \end{array}$$

$$(v_1 \text{ or } v_2 \text{ or } v_3 \text{ or } v_4)$$

Similarly

$$f(u_2) = v_4$$

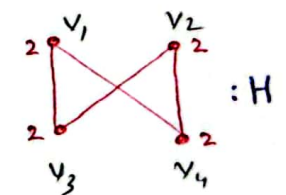
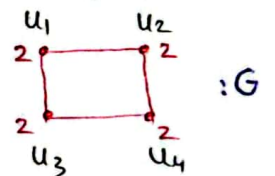
$$f(u_3) = v_3$$

$$f(u_4) = v_2$$

$$\deg(u_2) = 2 \begin{array}{l} 2 \\ 2 \end{array}$$

~~deg~~

$$(v_1 \mid v_2 \mid v_3 \mid v_4)$$



To see this one-to-one correspondence preserves adjacency relationship,

u_1, u_3 are adjacent in G , so there must be an edge between

$f(u_1) = v_1$ and $f(u_3) = v_3$. YES it is.

In the similar way you can check adjacency relationship for all $u \in U$.

The above graphs G and H are isomorphic.

NOTE:

There are $n!$ possible one-to-one Correspondence between vertex sets of two simple graphs with n vertices.

Testing each Such Correspondence to see whether it preserves adjacency and non-adjacency is Impractical if n is large.

Sometimes it is not hard to show that two graphs are not isomorphic.

As there is one-to-one Correspondence, so

- Isomorphic simple graphs must have **same # of vertices**
- Isomorphic simple graphs must have **same # of edges**

Also • degree of vertices in Isomorphic graphs must be the **same** (same degree sequence) i.e.

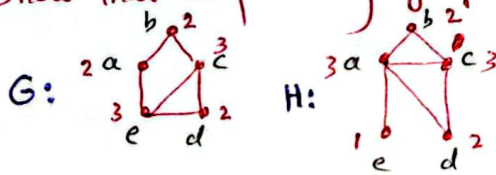
~~a vertex u in G is adjacent to v in G~~

a vertex u of degree d in G must Correspond to vertex $f(u)$ of degree d in H , because

a vertex u_1 in G is adjacent to u_2 iff $f(u_1)$ and $f(u_2)$ are adjacent in H .

Example #2

Show that the following graphs are not isomorphic.



Sol:-

1. Both have same # of vertices and edges

2. Degree Sequence: (increasing order)

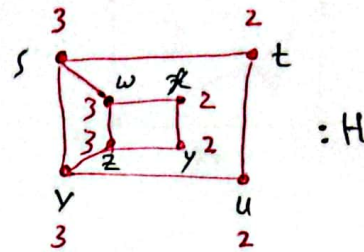
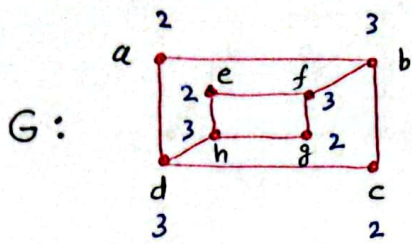
Degree Sequence of G : 3 3 2 2 2
Degree Sequence of H : 3 3 2 2 1 } $\neq$

So G and H are not isomorphic.

Example #3

Determine whether the graphs G and H are isomorphic.

Soln



1. Both have 8 vertices and 10 edges

Degree	# of vertices	
	G	H
2	4	4
3	4	4

3. $f(a) =$

deg(Adj)

$$\deg(a) = 2 \begin{cases} 3 \\ 3 \end{cases}$$

$$1. \deg(t) = 2 \begin{cases} 3 \\ 2 \end{cases} \quad \times$$

$$2. \deg(u) = 2 \begin{cases} 2 \\ 3 \end{cases} \quad \times$$

$$3. \deg(w) = 2 \begin{cases} 3 \\ 2 \end{cases} \quad \times$$

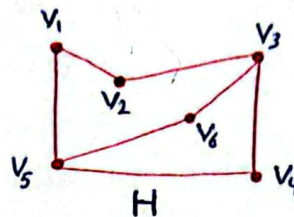
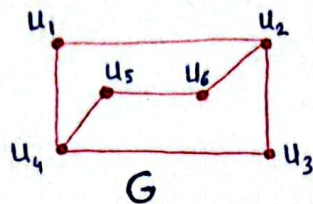
$$4. \deg(x) = 2 \begin{cases} 3 \\ 2 \end{cases} \quad \times$$

a must correspond to either t, u, x and u (as their deg is 2)

However, each of the four vertices in H is adjacent to another vertex of degree 2 in H which is not true for a in G (a is adjacent to vertices of degree 3.)

Example #4 :

Determine whether the graphs G and H show below are ISOMORPHIC -



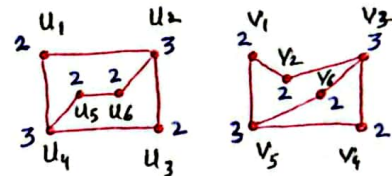
Sol:

1: Both graphs G and H have same # of vertices and same # of edges

	G	H
# of vertices :	6	6
# of Edges :	7	7

2: Both graphs G and H have same # of vertices of degree d

Degree	# of vertices	
	G	H
0	0	0
1	0	0
2	4	4
3	2	2



3. one-to-one Correspondence between the vertices of the two graphs that preserves the adjacency relationship.

$f(u_1) = ?$ $f(u_2) = ?$ $f(u_3) = ?$ $f(u_4) = ?$ $f(u_5) = ?$ $f(u_6) = ?$

$f(u_1) = v_6$

$\deg(u_1) = 2$ $\deg(v_6) = 2$: H

$\deg(v_6) = 2$: G

$f(u_2) = v_3$

$\deg(u_2) = 3$

1. $\deg(v_3) = 3$
2. $\deg(v_5) = 3$
(v_3 or v_5)

Choose anyone.

Important :

If choice of v_3 did not lead to isomorphism, we would not have established that G and H are not isomorphic, because second choice of v_5 may lead to isomorphism.

function f is called isomorphism if there exist one-to-one and onto function (bijection) f from U and V with the property that

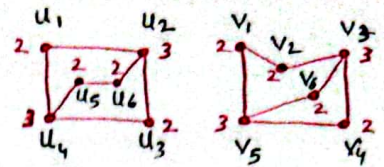
- a and b are adjacent in G iff $f(a)$ and $f(b)$ are adjacent in H for all values of a and b in U .

$$f(u_3) = v_4$$

$$\deg(u_3) = 2$$

1. $\deg(v_4) = 2$
2. $\deg(v_6) = 2$
- $\therefore (v_4 \text{ or } v_6)$

check anyone



$$f(u_4) = v_5$$

$$\deg(u_4) = 3$$

1. $\deg(v_3) = 3$
2. $\deg(v_5) = 3$
- $(v_3 \text{ or } v_5)$
- $\downarrow$
- (taken)

$$f(u_5) = v_1$$

$$\deg(u_5) = 2$$

1. $\deg(v_1) = 2$
2. $\deg(v_2) = 2$
- $\therefore (v_1 \text{ or } v_2)$

$$f(u_6) = v_2$$

$$\deg(u_6) = 2$$

1. $\deg(v_1) = 2$
2. $\deg(v_2) = 2$

$$f(u_1) = v_6$$

$$f(u_2) = v_3$$

$$f(u_3) = v_4$$

$$f(u_4) = v_5$$

$$f(u_5) = v_1$$

$$f(u_6) = v_2$$

vertices of
one-to-one Correspondence between G and H .

Now we need to show f preserves adjacency relationship i.e.,

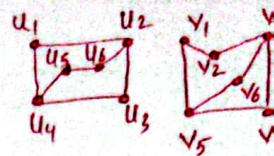
a and b are adjacent in G iff $f(a)$ and $f(b)$ are adjacent in H
for all a, b in U .

u_1 and u_4 are adjacent in G

$f(u_1)$ and $f(u_4)$ i.e. v_6 and v_5 are also adjacent in H .

one helpful way to do this is to use adjacency Matrices.

To show that f is an isomorphism, we can show that adjacency matrix G is the same as adjacency matrix H



$$\begin{aligned} f(u_1) &= v_6 \\ f(u_2) &= v_3 \\ f(u_3) &= v_4 \\ f(u_4) &= v_5 \\ f(u_5) &= v_1 \\ f(u_6) &= v_2 \end{aligned}$$

Adjacency Matrix for G :

$$A_G = \begin{matrix} & \begin{matrix} u_1 & u_2 & u_3 & u_4 & u_5 & u_6 \end{matrix} \\ \begin{matrix} u_1 \\ u_2 \\ u_3 \\ u_4 \\ u_5 \\ u_6 \end{matrix} & \begin{bmatrix} 0 & 1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 & 0 \end{bmatrix} \end{matrix}$$

A_G

Adjacency Matrix for H :

$$A_H = \begin{matrix} & \begin{matrix} v_6 & v_3 & v_4 & v_5 & v_1 & v_2 \end{matrix} \\ \begin{matrix} v_6 \\ v_3 \\ v_4 \\ v_5 \\ v_1 \\ v_2 \end{matrix} & \begin{bmatrix} 0 & 1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 & 0 \end{bmatrix} \end{matrix}$$

A_H

Because A_G and A_H are same, it follows that f preserves edges. we conclude that as f is an isomorphism, so G and H are isomorphic.

Note:-

If f turned out not to be an isomorphism, we would not have established that G and H are not isomorphic, because another correspondence of vertices (selection of v_5 ^{in case} ~~as~~ v_3 failed) as an image of u_2 in G and H may be an isomorphism.

$$A_G = \begin{matrix} & \begin{matrix} u_1 & u_2 & u_3 & u_4 & u_5 & u_6 \end{matrix} \\ \begin{matrix} u_1 \\ u_2 \\ u_3 \\ u_4 \\ u_5 \\ u_6 \end{matrix} & \begin{bmatrix} & & & & & \\ & & & & & 1 \\ & & & & & \\ & & & & & \\ & & & & & \\ & & & & & \\ & & & & & \end{bmatrix} \end{matrix}$$

u_2, u_6 is an edge of G
As $f(u_2) = v_3$
 $f(u_6) = v_2$

So v_3, v_2 must be an edge of H

$$A_H = \begin{matrix} & \begin{matrix} v_6 & v_3 & v_4 & v_5 & v_1 & v_2 \end{matrix} \\ \begin{matrix} v_6 \\ v_3 \\ v_4 \\ v_5 \\ v_1 \\ v_2 \end{matrix} & \begin{bmatrix} & & & & & \\ & & & & & 1 \\ & & & & & \\ & & & & & \\ & & & & & \\ & & & & & \\ & & & & & \end{bmatrix} \end{matrix}$$

image of u_2 $\rightarrow$ v_3
 $f(u_2) = v_3$
 $f(u_6) = v_2$

a and b : edge in G
i.e $a \neq b$ are adjacent
then $f(a)$ and $f(b)$ must be adjacent in H for all values of a and b .

All a and b are shown by adjacent matrix G and their images are shown by adjacent matrix H .